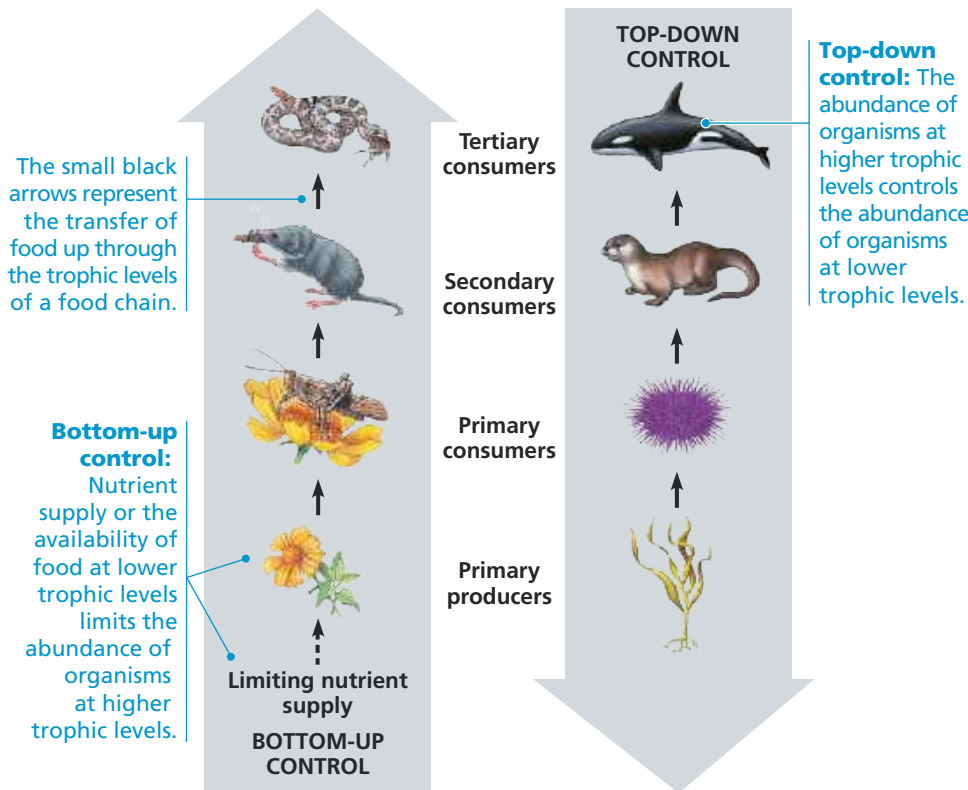


▼ **Figure 54.22 Bottom-up and top-down control.** The abundance of organisms at each trophic level can be controlled from the bottom up (by the availability of food or nutrients) or from the top down (by rates of consumption at higher trophic levels). Large gray arrows indicate the direction of control. Small black arrows show the transfer of food through a food chain.



availability of food at lower trophic levels. In this case, the supply of nutrients controls plant numbers, which control herbivore numbers, which in turn control predator numbers (**Figure 54.22**, left side). To change the structure of a bottom-up community, you would need to alter the biomass or abundance of organisms at lower trophic levels, allowing those changes to propagate up through the food web. For example, if you add nutrients to stimulate plant growth, then each of the higher trophic levels should also increase in biomass. If you change predator abundance, however, the effect should not extend down to the lower trophic levels.

In **top-down control**, the abundance of organisms at each trophic level is controlled by the abundance of consumers at higher trophic levels. Thus, in top-down control, predators limit herbivores, and herbivores limit plants (**Figure 54.22**, right side). In a lake community with four trophic levels, removing the top carnivores should increase the abundance of primary carnivores, decrease the number of herbivores, and increase phytoplankton abundance. The effects thus move down the trophic structure as alternating $\pm$ effects.

Ecologists have applied top-down control to improve water quality in lakes with a high abundance of algae. For example, in lakes with three trophic levels, removing fish should improve water quality by increasing zooplankton density, thereby decreasing algal populations (**Figure 54.23**). In lakes

with four trophic levels, adding top predators should have the same effect.

Ecologists in Finland used this approach to help purify Lake Vesijärvi, a large lake that was polluted with city sewage and industrial waste until 1976. After pollution controls reduced these inputs, the lake's water quality began to improve. In 1986, however, massive blooms of cyanobacteria began occurring. These blooms coincided with an increase in the population of roach, a fish species that eats zooplankton, which otherwise keep cyanobacteria in check. To reverse these changes, ecologists removed nearly a million kilograms of fish from the lake between 1989 and 1993, reducing roach abundance by about 80%. At the same time, they added a fourth trophic level by stocking the lake with pike perch, a predatory fish that eats roach. The water cleared up, and the last cyanobacterial bloom was in 1989. Ecologists continue to monitor the lake for cyanobacterial blooms and low oxygen

availability, but the lake has remained clear, even though roach removal ended in 1993.

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▼ **Figure 54.23 Restoring water quality in a lake with top-down control.** Decreasing the abundance of fish that ate zooplankton results in a decrease in the biomass of algae, improving water quality. Arrow thickness indicates the relative strength of each top-down control.

